Applications to statistics

December 2nd, 2025

How do statisticians describe variability?

Use a probability model for data.

Example:

Treat votes in a random sample of 500 people as 500 iid Bernoulli(p) random variables.

You did that on Quiz 12.

Two big questions

1. How do you know if a probability model is a good model for data?

Probability models can predict all sorts of things. All the voters in the sample could choose the same candidate just by chance. If the model can predict anything, how do we decide if it is good?

2. Given some data, how do you choose a probability model?

If 257 people in the sample say they will vote for the politician, what is it reasonable to say about the parameter p?

Question 1: testing a probability model

You were testing probability models in Homework 24.

To test, calculate a p-value, the probability that the model generates data with a "test statistic" like what you saw or more extreme.

Small p-value: model is unlikely to generate data like what you saw; that's evidence that the model you are testing does not fit some aspect of the data (determined by test statistic).

Question 2: choosing a model — the harder part

Choosing a model is complicated!

The harder decision is the form of the model.

For the sample of 500 voters:

- Independent
- Same probability p of picking our candidate (Bernoulli(p) for indicator)

MATH 313 is about all the choices for models and strategies for model selection.

Question 2: choosing a model — the easier part

Given the form of the model, determine parameters from data.

We settle for a “confidence interval.”

Example:

Sample of 500 voters.

Model votes with independent Bernoulli(p) random variables.

Given 257 “yes” votes, what should we pick for p?

Confidence intervals

257 “yes”: proportion $\hat{p}$ = 257/500 = 0.514.

In the model, $\hat{p} = (1/500)(X_1 + \dots + X_{500})$ is a random variable.

CLT says $\hat{p}$ is approximately normally distributed.

$E[\hat{p}] = p$

$\mathrm{Var}(\hat{p}) = (1/500)\, p\, (1 - p)$

$\sigma(\hat{p})$ = square root of $(1/500)\, p\, (1 - p)$ — called “standard error”

Confidence intervals continued

From the normal table,

P(–2 < standard normal < 2) is approximately 0.95.

There is a 95% chance that a normally distributed random variable is within 2 standard deviations of its expected value.

For $\hat{p}$, there is a 95% chance that $\hat{p}$ is within 2 standard errors of p.

In other words, there is a 95% chance that p is within 2 standard errors of $\hat{p}$.

Confidence intervals continued

$\sigma(\hat{p})$ = square root of (1/500) p (1 – p)

If we don't know p, we don't know the standard error $\sigma(\hat{p})$.

We estimate the standard error by replacing p with $\hat{p}$.

95% confidence interval:

[$\hat{p}$ – 2 (estimated standard error), $\hat{p}$ + 2 (estimated standard error)]

Confidence intervals concluded

The confidence interval is a random interval!

Probability that this interval contains the parameter p is (at least) 95%.

For 95% of data generated from the model, confidence interval contains p.

Informal meaning of a 95% interval:

- Assuming that the data come from independent Bernoulli(p) random variables
- and assuming that the sample is not unusual (not in extreme 5%),

the parameter p is in the confidence interval.

Mathematical statistics

Probability theory is the foundation of statistics.

Mathematical statistics uses probability theory to understand and to make precise what statisticians do.